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Science Fiction

APRIL 1944

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BY A. E. VAN VOGT

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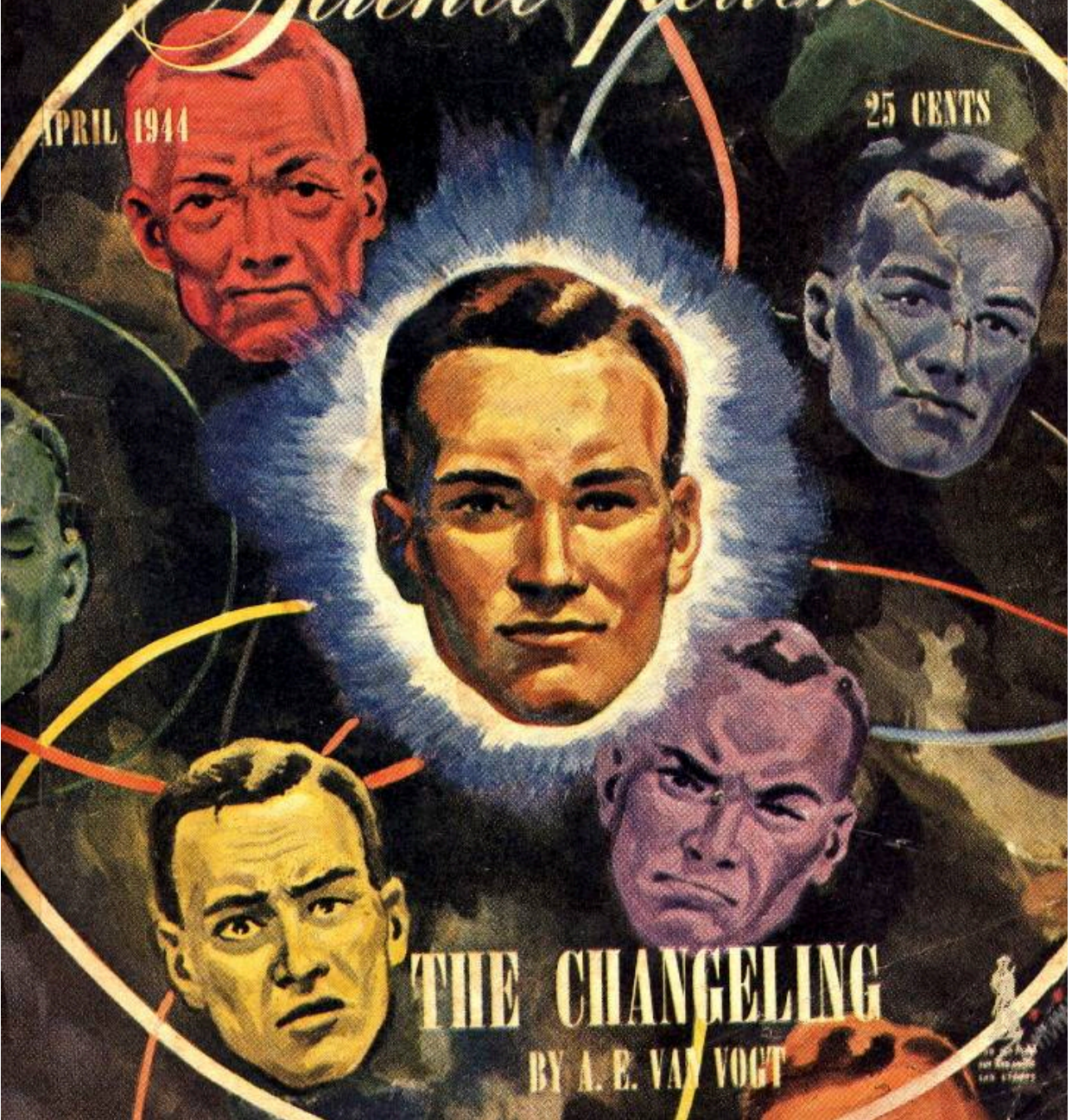
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Title: The Long Way

Author: George O. Smith

Illustrator: Frank Kramer

Release date: May 6, 2022 [eBook #68003]

Most recently updated: October 18, 2024

Language: English

Original publication: United States: Street & Smith Publications, Incorporated, 1944

Other information and formats: www.gutenberg.org/ebooks/68003

Credits: Greg Weeks, Mary Meehan and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK THE LONG WAY

The Long Way

By GEORGE O. SMITH

Illustrated by Kramer

**[Transcriber's Note: This etext was produced from
Astounding Science-Fiction, April 1944.
Extensive research did not uncover any evidence that
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Don Channing stood back and admired his latest acquisition with all of the fervency of a high school girl inspecting her first party dress. It was so apparent, this affection between man and gadget, that the workmen who were now carrying off the remnants of the packing case did so from the far side of the bench so that they would not come between the director of communications and the object of his affection. So intent was Channing in his adoration of the object that he did not hear the door open, nor the click of high heels against the plastic flooring. He was completely unaware of his surroundings until Arden said:

"Don, what off earth is that?"

"Ain't she a beaut?" breathed Channing.

"Jilted for a jimcrank," groaned Arden. "Tell me, my quondam husband, what is it?"

"Huh?" asked Don, coming to life once more.

"In plain, unvarnished words of one cylinder, what is that ... that, *that*?"

"Oh, you mean the transmission tube?"

"How do you do?" said Arden to the big tube. "Funny-looking thing, not like any transmitting tube I've ever seen before."

"Not a transmitting tube," explained Channing. "It is one of those power transmission tubes that Baler and Carroll found on the Martian desert."

"I presume that is why the etch says: 'Made by Terran Electric, Chicago'?"

Channing laughed. "Not one found—there was only one found. This is a carbon copy. They are going to revolutionize the transmission of power with 'em."

"Funny-looking gadget."

"Not so funny. Just alien."

"Know anything about it?"

"Not too much. But I've got Barney Carroll coming out here and a couple of guys from Terran Electric. I'm going to strain myself to keep from tinkering

with the thing until they get here."

"Can't you go ahead? It's not like you to wait."

"I know," said Channing. "But the Terran Electric boys have sewed up the rights to this dinkus so tight that it is squeaking. Seems to be some objection to working on them in the absence of their men."

"Why?"

"Probably because Terran Electric knows a good thing when they see it. Barney's latest 'gram said that they were very reluctant to rent this tube to us. Legally they couldn't refuse, but they know darned well that we're not going to run power in here from Terra—or anywhere else. They know we want it for experimentation, and they feel that it is their tube and that if any experimentation is going to take place, they're going to do it."

The workmen returned with two smaller cases; one of each they placed on benches to either side of the big tube. They knocked the boxes apart and there emerged two smaller editions of the center tube—and even Arden could see that these two were quite like the forward half and the latter half, respectively, of the larger tube.

"Did you buy 'em out?" she asked.

"No," said Don simply. "This merely makes a complete circuit."

"Explain that one, please."

"Sure. This one on the left is the input-terminal tube which they call the power-end. The good old D. C. goes in across these two terminals. It emerges from the big end, here, and bats across in a beam of intangible something-or-other until it gets to the relay tube where it is once more tossed across to the load-end tube. The power is taken from these terminals on the back end of the load-end tube and is then suitable for running motors, refrigerators, and so on. The total line-loss is slightly more than the old-fashioned transmission line. The cathode-dynode requires replacement about once a year. The advantages over high-tension wires are many; in spite of the slightly higher line-losses and the replacement trick, they are replacing long-lines everywhere.

"When they're properly aligned, they will scat right through a mountain of solid iron without attenuation. It takes one tower every hundred and seventy

miles, and the only restriction on tower height is that the tube must be above ground by ten to one the distance that could be flashed over under high intensity ultraviolet light."

"That isn't clear to me."

"Well, high-tension juice will flash over better under ultraviolet illumination. The tube must be high enough to exceed this distance by ten to one at the operating voltage of the stuff down the line. Another thing, the darned beam can be made to curve by adjusting the beam plates in the tube. The boys in the Palanortis Jungles say they're a godsend, since there are a lot of places where the high-tension towers would be impossible since the Palanortis Whitewood grows about a thousand feet tall."

"You'd cut a lot of wood to ream a path through from Northern Landing to the power station on the Boiling River," said Arden.

"Yeah," drawled Don, "and towers a couple of hundred miles apart are better than two thousand feet. Yeah, these things are the nuts for getting power shipped across country."

"Couldn't we squirt it out from Terra?" asked Arden. "That would take the curse off of our operating expenses."

"It sure would," agreed Channing heartily. "But think of the trouble in aligning a beam of that distance. I don't know—there's this two hundred mile restriction, you know. They don't transmit worth a hoot over that distance, and it would be utterly impossible to maintain stations in space a couple of hundred miles apart, even from Venus, from which we maintain a fairly close tolerance. We might try a hooting big one, but the trouble is that misalignment of the things results in terrible effects."

The door opened and Charley Thomas and Walt Franks entered.

"How's our playthings?" asked Walt.

"Cockeyed looking gadgets," commented Charley.

"Take a good look at 'em," said Channing. "Might make some working X-ray plates, too. It was a lucky day that these got here before the boys from Terran

Electric. I doubt that they'd permit that."

"O.K.," said Charley. "I'll bring the X-ray up here and make some pix. You'll want working prints; Walton will have to take 'em and hang dimensions on to fit."

"And we," said Channing to Walt Franks, "will go to our respective offices and wait until the Terran Electric representatives get here."

The ship that came with the tubes took off from the landing stage, and as it passed their observation dome, it caught Don's eye. "There goes our project for the week," he said.

"Huh?" asked Walt.

"He's been like that ever since we tracked him down with the *Relay Girl*," said Arden.

"I mean the detection of driver radiation," said Channing.

"Project for the week?" asked Walt. "Brother, we've been tinkering with that idea for months, now."

"Well," said Don, "there goes four drivers, all batting out umpty-ump begawatts of something. They can hang a couple of G on a six-hundred foot hull for hours and hours. The radiation they emit must be detectable; don't tell me that such power is not."

"The interplanetary companies have been tinkering with drivers for years and years," said Walt. "They have never detected it?"

"Could be, but there are a couple of facts that I'd like to point out. One is that they're not interested in detection. They only want the best in driver efficiency. Another thing is that the radiation from the drivers is sufficient to ionize atmosphere into a dull red glow that persists for several minutes. Next item is the fact that we on Venus Equilateral should be able to invent a detector; we've been tinkering with detectors long enough. Oh, I'll admit that it is secondary-electronics—"

"Huh? That's a new one on me."

"It isn't electronics," said Channing. "It's subetheric or something like that. We'll call it sub-electronics for lack of anything else. But we should be able to detect it somehow."

"Suppose there is nothing to detect?"

"That smacks of one hundred percent efficiency," laughed Don. "Impossible."

"How about an electric heater?" asked Arden.

"Oh Lord, Arden, an electric heater is the most ineffic—"

"Is it?" interrupted Arden with a smile. "What happens to radiation when intercepted?"

"Turns to heat, of course."

"That takes care of the radiation output," said Arden. "Now, how about electrical losses?"

"Also heat."

"Then everything that goes into an electric heater emerges as heat," said Arden.

"I get it," laughed Walt. "Efficiency depends upon what you hope to get. If what you're wanting is losses, anything that is a total loss is one hundred percent efficient. Set your machine up to waste power and it becomes one hundred percent efficient as long as there is nothing coming from the machine that doesn't count as waste."

"Fine point for argument," smiled Channing. "But anything that will make atmosphere glow that dull red after the passage of a ship will have enough waste to detect. Don't tell me that the red glow enhances the drive."

The door opened again and Charley came in with a crew of men. They ignored the three, and started to hang heavy cloth around the walls and ceiling. Charley watched the installation of the barrier-cloth and then said: "Beat it—if you want any young Channings!"

Arden, at least, had the grace to blush.

The tall, slender man handed Don an envelope full of credentials. "I'm Wesley Farrell," he said. "Glad to have a chance to work out here with you fellows."

"Glad to have you," said Don. He looked at the other man.

"This is Mark Kingman."

"How do you do?" said Channing. Kingman did not impress Channing as being a person whose presence in a gathering would be demanded with gracious shouts of glee.

"Mr. Kingman is an attorney for Terran Electric," explained Wesley.

Kingman's pedestal was lowered by Channing.

"My purpose," said Kingman, "is to represent my company's interest in the transmission tube."

"In what way?" asked Don.

"Messrs. Baler and Carroll sold their discovery to Terran Electric outright. We have an iron-bound patent on the device and/or any developments of the device. We hold absolute control over the transmission tube, and therefore may dictate all terms on which it is to be used."

"I understand. You know, of course, that our interest in the transmission tube is purely academic."

"I have been told that. We're not too certain that we approve. Our laboratories are capable of any investigation you may desire, and we prefer that such investigations be conducted under our supervision."

"We are not going to encroach on your power rights," explained Channing.

"Naturally," said Kingman in a parsimonious manner. "But should you develop a new use for the device, we shall have to demand that we have complete rights."

"Isn't that a bit high-handed?" asked Don.

"We think not. It is our right."

"You're trained technically?" asked Don.

"Not at all. I am a lawyer, not an engineer. Mr. Farrell will take care of the technical aspects of the device."

"And in looking out for your interests, what will you require?"

"Daily reports from your group. Daily conferences with your legal department. These reports should be prepared prior to the day's work so that I may discuss with the legal department the right of Terran Electric to permit or to disapprove the acts."

"You understand that there may be a lot of times when something discovered at ten o'clock may change the entire program by ten oh six?"

"That may be," said Kingman, "but my original statements must be adhered to, otherwise I am authorized to remove the devices from your possession. I will go this far, however; if you discover something that will change your program for the day, I will then call an immediate conference which should hurry your program instead of waiting until the following morning for the decision."

"Thanks," said Channing dryly. "First, may we take X-ray prints of the devices?"

"No. Terran Electric will furnish you with blueprints which we consider suitable." Kingman paused for a moment. "I shall expect the complete program of tomorrow's experiments by five o'clock this evening."

Kingman left, and Wes Farrell smiled uncertainly. "Shall we begin making the list?"

"Might as well," said Channing. "But, how do you lay out a complete experimental program for twelve hours ahead?"

"It's a new one on me, too," said Farrell.

"Well, come on. I'll get Walt Franks, and we'll begin."

"I wonder if it might not be desirable for Kingman to sit in on these program-settings?" said Channing, after a moment of staring at the page before him.

"I suggested that to him. He said 'No'. He prefers his information in writing."

Walt came in on the last words. Channing brought Franks up to date and Walt said: "But why would he want a written program if he's going to disallow certain ideas?"

"Sounds to me like he's perfectly willing to let us suggest certain lines of endeavor; he may decide that they look good enough to have the Terran

Electric labs try themselves," said Channing.

Wes Farrell looked uncomfortable.

"I have half a notion to toss him out," Channing told Farrell. "I also have half a notion to make miniatures of this tube and go ahead and work regardless of Kingman or Terran Electric. O.K., Wes, we won't do anything illegal. We'll begin by making our list."

"What is your intention?" asked Wes.

"We hope that these tubes will enable us to detect driver radiation, which will ultimately permit us to open ship-to-ship two-way communication."

"May I ask how you hope to do this?"

"Sure. We're going to cut and try. No one knows a thing about the level of driver-energy; we've assigned a selected name for it: Subelectronics. The driver tube is akin to this transmission tube, if what I've been able to collect on the subject is authentic. By using the transmission tube—"

"Your belief is interesting. I've failed to see any connection between our tube and the driver tube."

"Oh sure," said Channing expansively. "I'll admit that the similarity is of the same order as the similarity between an incandescent lamp and a ten dynode, electron-multiplier such as we use in our final beam stages. But recall this business of the cathode-dynode. In both, the emitting surface is bombarded by electrons from electron guns. They both require changing."

"I know that, but the driver cathode disintegrates at a rate of loss that is terrific compared to the loss of emitting surface in the transmission tube."

"The driver cathode is worth about two hundred G-hours. But remember, there is no input to the driver such as you have in the transmission tube. The power from the driver comes from the disintegration of the cathode surface—there isn't a ten thousandth of an inch of plating on the inside of the tube to show where it went. But the transmission tube has an input and the tube itself merely transduces this power to some level of radiation for transmission. It is re-transduced again for use. But the thing is this: Your tube is the only thing we know of that will accept subelectronic energy and use it. If the driver and the transmission tubes are similar in operational spectrum, we may be able to detect driver radiation by some modification."

"That sounds interesting," said Wes. "I'll be darned glad to give you a lift."

"Isn't that beyond your job?" asked Channing.

"Yeah," drawled Farrell, "but could you stand by and watch me work on a beam transmitter?"

"No—"

"Then don't expect me to watch without getting my fingers dirty," said Farrell cheerfully. "Sitting around in a place like this would drive me nuts without something to do."

"O.K., then," smiled Don. "We'll start off by building about a dozen miniatures. We'll make 'em about six inches long—we're not going to handle much power, you know. That's first."

Kingman viewed the list with distaste. "There are a number of items here which I may not allow," he said.

"For instance?" asked Channing with lifted eyebrows.

"One, the manufacture or fabrication of power transmission tubes by anyone except Terran Electric is forbidden. Two, your purpose in wanting to make tubes is not clearly set forth. Three, the circuits in which you intend to use these tubes is unorthodox, and must be clearly and fully drawn and listed."

"Oh spinach! How can we list and draw a circuit that is still in the embryonic stage?"

"Then clarify it. Until then I shall withhold permission."

"But look, Mr. Kingman, we're going to develop this circuit as we go along."

"You mean that you are going to fumble your way through this investigation?"

"We do not consider a cut-and-try program as fumbling," said Walt Franks.

"I am beginning to believe that your research department has not the ability to reduce your problems to a precise science," said Kingman scornfully.

"Name me a precise science," snapped Channing, "or even a precise art!"

"The legal trade is as precise as any. Everything we do is done according to legal precedent."

"I see. And when there is no precedent?"

"Then we all decide upon the proper course, and establish a precedent."

"But I've got to show you a complete circuit before you'll permit me to go ahead?"

"That's not all. Your program must not include reproducing these tubes either in miniature or full size—or larger. Give me your requirements and I shall request Terran Electric to perform the fabrication—"

"Look, Kingman, Venus Equilateral has facilities to build as good a tube as Terran Electric. I might even say better, since our business includes the use, maintenance, and development of radio tubes; your tubes are not too different from ours. Plus the fact that we can whack out six in one day, whilst it will take seventy-three hours to get 'em here after they're built on Terra."

"I'm sorry, but the legal meaning of the patent is clear. Where is your legal department?"

"We have three. One on each of the Inner Planets."

"I'll request you to have a legal representative come to the Station so that I may confer with him. One with power of attorney to act for you."

"Sorry," said Channing coldly. "I wouldn't permit any attorney to act without my supervision."

"That's rather a backward attitude," said Kingman. "I shall still insist on conducting my business with one of legal mind."

"O.K. We'll have Peterman come out from Terra. But he'll still be under my supervision."

"As you wish. I may still exert my prerogative and remove the tubes from your possession."

"You may find that hard to do," said Channing.

"That's illegal!"

"Oh no, it won't be. You may enter the laboratory at any time and remove the tubes. Of course, if you are without technical training you may find it most

difficult to disconnect the tubes without getting across a few thousand volts. That might be uncomfortable."

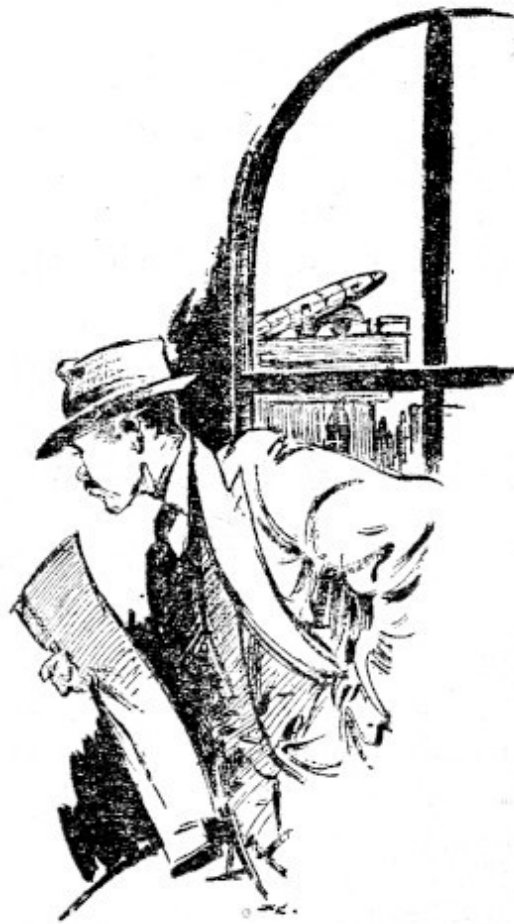
"Are you threatening me?" said Kingman, bristling. His stocky frame didn't take to bristling very well, and he lost considerable prestige in the act.

"Not at all. I'm just issuing a fair warning that the signs that say: DANGER! HIGH VOLTAGE! are not there for appearance."

"Sounds like a threat to me."

"Have I threatened you? It sounds to me as though I were more than anxious for your welfare. Any threat of which you speak is utterly without grounds, and is a figment of your imagination; based upon distrust of the Interplanetary Communications Company, and the personnel of the Venus Equilateral Relay Station."

Kingman shut up. He went down the list, marking off items here and there. While he was marking, Channing scribbled a circuit and listed the parts. He handed it over as Kingman finished.



"This is your circuit?" asked the lawyer skeptically.

"Yes."

"I shall have to ask for an explanation of the symbols involved."

"I shall be happy to present you with a book on essential radio technique," offered Channing. "A perusal of which will place you in possession of considerable knowledge. Will that suffice?"

"I believe so. I can not understand how; being uncertain of your steps a few minutes ago, you are now presenting me with a circuit of your intended experiment."

"The circuit is, of course, merely symbolic. We shall change many of the constants before the day is over—in fact, we may even change the circuit."

"IT shall require a notice before each change so that I may pass upon the legal aspects."

"Walt," said Don, "will you accompany me to a transparency experiment on the Ninth Level?"

"Be more than glad to," said Walt. "Let's go!"

They left the office quickly, and started for Joe's. They had not reached the combined liquor-vending and restaurant establishment when the communicator called for Channing. It was announcing the arrival of Barney Carroll, so instead of heading for Joe's, they went to the landing stage at the south end of the Station to greet the visitor.

"Barney," said Don, "of all the companies, why did you pick on Terran Electric?"

"Gave us the best deal," said the huge, grinning man.

"Yeah, and they're getting the best of my goat right now."

"Well, Jim and I couldn't handle anything as big as the power transmission set-up. They paid out a large slice of jack for the complete rights. All of us are well paid now. After all, I'm primarily interested in Martian artifacts, you know."

"I wonder if they had lawyers," smiled Walt wryly.

"Probably. And, no doubt, the legals had a lot to do with the fall of the Martian Civilization."

"As it will probably get this one so bound up with red tape that progress will be impossible—or impractical."

"Well, Barney, let's take a run up to the lab. We can make paper-talk even if Brother Kingman won't let us set it to soldering iron. There are a lot of things I want to ask you about the tube."

They sat around a drawing table and Channing began to sketch. "What I'd hoped to do is this," he said, drawing a schematic diagram. "We're not interested in power transmission, but your gadget will do a bit of voltage amplification because of its utter indifference to the power-line problem of impedance matching. We can take a relay tube and put in ten watts, say, across ten thousand ohms. That means the input will be somewhat above three hundred volts. Now, if our output is raced across a hundred thousand ohms, ten watts will give us one thousand volts. So we can get voltage amplification at the expense of current—which we will not need. Unfortunately, the relay tube as well as the rest of the system will give out with the same kind of power that it is impressed with—so we'll have amplification of driver radiation. Then we'll need a detector. We haven't been able to get either yet, but this is a start, providing that Terran Electric will permit us to take a deep breath without wanting to pass on it."

"I think you may be able to get amplification," said Barney. "But to do it, you'll have to detect it first."

"Huh?"

"Sure. Before these darned things will work, this in-phase anode must be right on the beam. That means that you'll require a feed-back circuit from the final stage to feed the in-phase anodes. Could be done without detection, I suppose."

"Well, for one thing, we're going to get some amplification if we change the primary anode—so. That won't permit the thing to handle any power, but it will isolate the output from the input and permit more amplification. Follow?"

"Can we try it?"

"As soon as I get Terran Electric's permission."

"Here we go again!" groaned Walt.

"Yeah," said Don to Barney, "now you'll see the kind of birds you sold your gadget to."

They found Kingman and Farrell in conference. Channing offered his suggestion immediately, and Kingman looked it over, shaking his head.

"It is not permitted to alter, change, rework, or repair tubes owned by Terran Electric," he said.

"What are we permitted to do?" asked Channing.

"Give me your recommendation and I shall have the shop at Terran Electric perform the operation."

"At cost?"

"Cost plus a slight profit. Terran Electric, just as Communications, is not in business from an altruistic standpoint."

"I see."

"Also," said Kingman severely, "I noticed one of your men changing the circuit slightly without permission. Why?"

"Who was it?"

"The man known as Thomas."

"Charley Thomas is in charge of development work," said Channing. "He probably noticed some slight effect that he wanted to check."

"He should have notified me first—I don't care how minute the change. I must pass on changes first."

"But you wouldn't know their worth," objected Barney.

"No, but Mr. Farrell does, and will so advise me."

Wes looked at Channing. "Have you been to the Ninth Level yet?"

"Nope," said Channing.

"May I accompany you?"

Channing looked at Farrell critically. The Terran Electric engineer seemed sincere, and the pained expression on his face looked like frustrated sympathy to Don. "Come along," he said.

Barney smiled cheerfully at the sign on Joe's door. "That's a good one, 'Best Bar in Twenty-seven Million Miles, Minimum!' What's the qualification

for?"

"That's about as close as Terra gets. Most of the time the nearest bar is at Northern Landing, Venus; sixty-seven million miles from here. Come on in and we'll get plastered."

Farrell said: "Look, fellows, I know how you feel. They didn't tell me that you weren't going to be given permission to work. I understood that I was to sort of walk along, offer suggestions, and sort of prepare myself to take over some research myself. This is sickening."

"I think you mean that."

"May I use your telephone. I want to resign."

"Wait a minute. If you're that sincere, why don't we outguess 'em?"

"Could do," said Wes. "How?"

"Is there any reason why we can't take a poke to Sol himself?"

"You mean haul power out of the sun?"

"That's the general idea. Barney, what do you think?"

"Could be—but it would take a redesign."

"Fine. And may we pray that the redesign is good enough to make a difference to the Interplanetary Patent Office." Channing called Joe. "The same. Three Moons all around. Scotch," he explained to the others, "synthesized in the Palanortis Country."

"Our favorite import," said Walt.

Joe grinned. "Another tablecloth session in progress?"

"Could be. As soon as we oil the think-tank, we'll know for sure."

"What does he mean?" asked Barney.

Joe smiled. "They all have laboratories and draftsmen and textbooks," he said. "But for real engineering, they use my tablecloths. Three more problems and I'll have a complete tablecloth course in astrophysics, with a sideling in cartooning, and a minor degree in mechanical engineering."

"Oh?"

"Sure. Give 'em free hand, and a couple of your tubes and a tablecloth and they'll have 'em frying eggs by morning. When I came out here, they demanded a commercial bond and I thought they were nuts. Who ever heard of making a restaurateur post a bond? I discovered that all of their inventions are initially tinkered out right here in the dining room—I could steal 'em blind if I were dishonest!" Joe smiled hugely. "This is the only place in the system where the tablecloths have been through blueprint machines. That," he said confidentially to Barney, "is why some of the stuff is slightly garbled. Scotch mixed with the drawings. They have the cloths inspected by the engineering department before they're laundered; I lose a lot of tablecloths that way."

Joe left cheerfully amid laughter.

The Three Moons came next, and then Don began to sketch. "Suppose we make a driver tube like this," he said. "And we couple the top end, where the cathode is to the input side of the relay tube. Only the input side will require a variable-impedance anode, coupled back from the cathode to limit the input to the required value. Then the coupling anodes must be served with an automatic-coupling circuit so that the limiting power is passed without wastage."

Barney pulled out a pencil. "If you make that automatic-coupling circuit dependent upon the output from the terminal ends," he said, "it will accept only the amount of input that is required by the power being used from the output. Over-cooling these two anodes will inhibit the power-intake."

"Right," said Wes. "And I am of the opinion that the power available from Sol is of a magnitude that will permit operation over and above the limit."

"Four million tons of energy per second!" exploded Walt. "That's playing with fire!"

"You bet. We'll fix 'em with that!"

"Our experience with relay tubes," said Farrell slowly, "indicates that some increase in range is possible with additional anode-focusing. Build your tube-top with an extra set of anodes, and that'll give us better control of the beam."

"We're getting farther and farther from the subject of communication," said Channing with a smile. "But I think that we'll get more out of this."

"How so?"

"Until we get a chance to tinker with those tubes, we won't get ship-to-ship two ways. So we'll gadgeteer up something that will make Terran Electric foam at the mouth, and swap a hunk of it for full freedom in our investigations. Or should we bust Terran Electric wholeheartedly?"

"Let's slug 'em," said Walt.

"Go ahead," said Wes. "I'm utterly disgusted, though I think our trouble is due to the management of Terran Electric. They like legal tangles too much."

"We'll give 'em a legal tangle," said Barney. He was adding circuits to the tablecloth sketch.

Channing, on his side, was sketching in some equations, and Walt was working out some mechanical details. Joe came over, looked at the tablecloth, and forthright went to the telephone and called Walton. The mechanical designer came, and Channing looked up in surprise. "Hi," he said. "I was about to call you."

"Joe did."

"O.K. Look, Ted, can you fake us up a gadget like this?"

Walton looked the thing over. "Give me about ten hours," he said.

"We've got a spare turnover driver from the *Relay Girl* that we can hand-carve. There are a couple of water-boilers that we can strip, cut open, and make to serve as the top end. How're you hoping to maintain the vacuum?"

"Yes," said Wes Farrell, "That's going to be the problem. If there's any adjusting of electrodes to do, this'll take months."

"That's why we, on Venus Equilateral, are ahead of the whole ding-busted solar system in tube development," said Don. "We'll run the thing out in the open—and I *do* mean open! Instead of the tube having the insides exhausted, the operators will have their envelopes served with fresh, canned air."

"Like a cartoon I saw somewhere," grinned Walt. "Had a bird in full armor tinkering with a radio set. The caption was: 'Why shield the set!'"

"Phooey," said Ted Walton, "Look, Tom Swift, is this another one of the Franks' brainchildren?"

"Tom Swift?" asked Wes.

"Yeah. That's the nom de plume he invents under. The other guy we call Captain Lightning."

"Oh?" asked Farrell, "Do you read him, too?"

"Sure," grinned Walton. "And say, speaking of comics, I came upon an old, old volume of Webster's International Dictionary in a rare-edition library a couple of months ago in Chicago, and they define 'Comic' as amusing, funny, and ludicrous; not imaginative fiction. How things change."

"They do."

"But to get back to this gold-berg, what is it?"

"Ted," said Channing soberly, "sit down!" Walton did. "Now," grinned Channing, "this screwball gadget is an idea whereby we hope to draw power out of the sun."

Walton swallowed once, and then waved for Joe. "Double," he told the restaurateur. Then to the others he said, "Thanks for seating me. I'm ill, I think. Hearing things. I could swear I heard someone say that this thing is to take power from Sol."

"That's it."

"Um-m-m. Remind me to quit Saturday. This is no job for a man beset by hallucinations."

"You grinning idiot, we're not fooling."

"Then you'd better quit," Walton told Don. "This is no job for a bird with delusions of grandeur, either. Look, Don, you'll want this in the experimental blister at South end? On a coupler to the beam-turret so that it'll maintain direction at Sol?"

"Right. Couple it to the rotating stage if you can. Remember, that's three miles from the South end."

"We've still got a few high-power selsyns," said Walton, making some notations of his own on the tablecloth. "And thanks to the guys who laid out

this Station some years ago, we've plenty of unused circuits from one end to the other. We'll couple it, all right. Oh mother. Seems to me like you got a long way off of your intended subject. Didn't you start out to make a detector for driver radiation?"

"Yup."

"And you end up tapping the sun. D'ye think it'll ever replace slave labor?"

"Could be. Might even replace the coal mine. That's to be seen. Have any idea of how long you'll be?"

"Make it ten hours. I'll get the whole crew on it at once."

"Fine."

"But look. What's the reason for this change in program?"

"That's easy," said Don. "First, we had a jam session. Second, we've come to the conclusion that the longest way around is often the shortest way home. We're now in the throes of building something with which to dazzle the bright-minded management of Terran Electric and thus make them susceptible to our charm. We want a free hand at the transmission tubes, and this looks like a fair bit of bait."

"I get it. Quote: 'Why buy power from Terran Electric? Hang a Channing Power Beam on your chimney pot and tap the sun!' Whoa, Maizie. Bring on the needle, Watson. Hang out the flags, fire the cannon, ring the bells; for Venus Equilateral is about to hang a pipeline right into four million tons of energy per second! Don, that's a right, smart bit of power to doodle with. Can you handle it?"

"Sure," said Channing with a wave of his hand, "we'll hang a fuse in the line!"

"O.K.," said Walton, sweeping the tablecloth off the table like Mysto, the Magician; right out from under the glasses, "I'll be back—wearing my asbestos pants!"

Wes Farrell looked dreamily at the ceiling. "This *is* a screwy joint," he said idly. "What do we do for the next ten hours?"

"Red Herring stuff," said Channing with what he hoped was a Machiavellian leer.

"Such as?"

"Making wise moves with the transmission tubes. Glomming the barrister's desk with proposed ideas for his approval; as many as we can think of so that he'll be kept busy. We might even think of something that may work, meanwhile. Come, fellow conspirators, to horse!" Channing picked up his glass and drained it, making a wry face. "Rotten stuff—I wish I had a barrel of it!"

Channing surveyed the set-up in the blister. He inspected it carefully, as did the others. When he spoke, his voice came through the helmet receivers with a slightly tinny sound: "Anything wrong? Looks O.K. to me."



"O.K. by me, too," said Farrell.

"Working in suit is not the best," said Don. "Barney, you're the bright-eyed lad, can you align the plates?"

"I think so," came the muffled booming of Barney's powerful voice. "Gimme screwdriver!"

Barney fiddled with the plate-controls for several minutes. "She's running on dead center alignment, now," he announced.

"Question," put in Wes, "do we get power immediately, or must we wait whilst the beam gets there and returns?"

"You must run your power line before you get power," said Walt. "My money is on the wait."

"Don't crack your anode-coupling circuit until then," warned Wes. "We don't know a thing about this; I'd prefer to let it in easy-like instead of opening the gate and letting the whole four million tons per second come roaring in through this ammeter."

"Might be a little warm having Sol in here with us," laughed Channing. "This is once in my life when we don't need a milliammeter, but a million-ammeter!"

"Shall we assign a pseudonym for it?" chuckled Walt.

"Let's wait until we see how it works."

The minutes passed slowly, and then Wes announced: "She should be here. Crack your anode-coupler, Barney."

Barney advanced the dial, gingerly. The air that could have grown tense was, of course, not present in the blister. But the term is but a figure of speech, and therefore it may be proper to say that the air grew tense. Fact is, it was the nerves of the men that grew tense. Higher and higher went the dial, and still the meter stayed inert against the zero-end pin.

"Not a wiggle," said Barney in disgust. He twirled the dial all the way around, and snorted. The meter left the zero pin ever so slightly.

Channing turned the switch that increased the sensitivity of the meter until the needle stood halfway up the scale.

"Solar power, here we come," he said in a dry voice. "One half ampere at seven volts! Three and one half watts. Bring on your atom-smashers! Bring on your power-consuming factory-districts. Hang the whole load of Central United States on the wires, for we have three and one half watts! Just enough to run an electric clock!"

"But would it keep time?" asked Barney. "Is the frequency right?"

"Nope—but we'd run it. Look, fellows, when anyone tells you about this, insist that we got thirty-five hundred milliwatts on our first try. It sounds bigger."

"O.K., so we're getting from Sol just about three tenths of the soup we need to make the set-up self-sustaining," said Walt. "Wes, this in-phase anode of yours—what can we do with it?"

"If this thing worked, I was going to suggest that there is enough power out there to spare. We could possibly modulate the in-phase anode with anything we wanted, and there would be enough junk floating around in the photosphere to slam on through."

"Maybe it is that lack of selectivity that licks us now," said Don. "Run the voltage up and down a bit. There should be D.C. running around in Sol, too."

"Whatever this power-level is running at," said Barney, "we may get in-phase voltage—or in-phase power by running a line from the power terminal back. Move over, boys, I'm going to hang a test clip in here."

Barney's gloved hands fumbled a bit, but the clip was attached. He opened the anode-coupler once again, and the meter slammed against the full-scale peg.

"See?" he said triumphantly.

"Yup," said Channing cryptically. "You, Bernard, have doubled our input."

"Mind if I take a whack at aligning it?" asked Wes.

"Go ahead. What we need is a guy with eyes in his fingertips. Have you?"

"No, but I'd like to try."

Farrell worked with the deflection plate alignment, and then said, ruefully: "No dice. Barney had it right on the beam."

"Is she aligned with Sol?" asked Channing.

Walt squinted down the tube. "Couldn't be better," he said, blinking.

"Could it be that we're actually missing Sol?" asked Don. "I mean, could it be that line-of-sight and line-of-power aren't one and the same thing?"

"Could be," acknowledged Wes. Walt stepped to the verniers and swung the big intake tube over a minute arc. The meter jumped once more, and Channing stepped the sensitivity down again. Walt fiddled until the meter read maximum and then he left the tube that way.

"Coming up," said Channing. "We're now four times our original try. We now have enough juice to run an electric train—a toy train! Someone think of something else, please. I've had my idea for the day."

"Let's juggle electrode-spacing," suggested Wes.

"Can do," said Walt, brandishing a huge spanner wrench in one gloved hand.

Four solid, futile hours later, the power output of the solar beam was still standing at a terrifying fourteen watts. Channing was scratching furiously on a pad of paper with a large pencil; Walt was trying voltage-variations on the supply-anodes in a desultory manner; Barney was measuring the electrode spacing with a huge vernier rule, and Wes was staring at the sun, dimmed to seeable brightness by a set of dark glasses.

Wes was muttering to himself. "Electrode-voltages, O.K. ... alignment perfect ... solar power output ... not like power-line electricity ... solar composition ... Russell's Mixture—"

"Whoooo said that!" roared Channing.

"Who said what?" asked Barney.

"Why bust our eardrums?" objected Walt.

"What do you mean?" asked Wes, coming to life for the moment.

"Something about Russell's Mixture. Who said that?"

"I did. Why?"

"Look, Wes, what are your cathodes made of?"

"Thorium, C. P. metal. That's why they are shipped in metal containers in a vacuum."

"What happens if you try to use something else?"

"Don't work very well. In fact, if the output cathode and the input dynode are not the same metal, they won't pass power at all."

"You're on the trail right now!" shouted Channing. "Russell's Mixture?"

"Sounds like a brand of smoking tobacco to me. Mind making a noise like an encyclopedia and telling me what is Russell's Mixture?"

"Russell's Mixture is a conglomeration of elements which go into the making of Sol—and all the other stars," explained Don. "Hydrogen, Oxygen, Sodium, and Magnesium, Iron, Silicon, Potassium, and Calcium. They, when mixed according to the formula for Russell's Mixture, which can be found in any book on the composition of stars, become the most probable mixture of metals. They—Russell's Mixture—go into the composition of all stars, what isn't mentioned in the mix isn't important."

"And what has this Russell got that we haven't got?" asked Walt.

"H, O, Na, Mg, Fe, Si, K, and Ca. And we, dear people, have Th, which Russell has not. Walt, call the metallurgical lab and have 'em whip up a batch."

"Cook to a fine edge and serve with a spray of parsley? Or do we cut it into cubes—"

"Go ahead," said Channing. "Be funny. You just heard the man say that dissimilar dynode-cathodes do not work. What we need for our solar beam is a dynode of Russell's Mixture so that it will be similar to our cathode—which in this case is Sol. Follow me?"

"Yeah," said Walt. "I follow, but brother I'm a long way behind. But I'll catch up," he promised as he made connection between his suit-radio and the Station communicator system. "Riley," he said, "Here we go again. Can you whip us up a batch of Russell's Mixture?"

Riley's laugh was audible to the others, since it was broadcast by Walt's set. "Yeah, man, we can—if it's got metal in it? What, pray tell, is Russell's Mixture?"

Walt explained the relation between Russell's Mixture and the composition of Sol.

"Sun makers, hey?" asked Riley. "Is the chief screwball there?"

"Yup," said Walt, grinning at Don.

"Sounds like him. Yeah, we can make you an alloy consisting of Russell's Mixture. Tony's got it here, now, and it doesn't look hard. How big a dynode do you want?"

Walt gave him the dimensions of the dynode in the solar tube.

"Cinch," said Riley. "You can have it in two hours."

"Swell."

"But it'll be hotter than hell. Better make that six or seven hours. We may run into trouble making it jell."

"I'll have Arden slip you some pectin," said Walt. "Tomorrow morning then?"

"Better. That's a promise."

Walt turned to the rest. "If any of us can sleep," he said, "I'd suggest it. Something tells me that tomorrow is going to be one of those days that mother told me about. I'll buy a drink."

Walt opened the anode-coupler circuit, and the needle of the output ammeter slammed across the scale and wound the needle halfway around the stop pin. The shunt, which was an external, high-dissipation job, turned red, burned the paint off of its radiator fins, and then proceeded to melt. It sputtered in flying droplets of molten metal. Smoke spewed from the case of the ammeter, dissipating in the vacuum of the blister.

Walt closed the coupler circuit.

"Whammo!" he said. "Mind blowing a hundred-amp meter?"

"No," grinned Don. "I have a thousand amp job that I'll sacrifice in the same happy-hearted fashion. Get an idea of the power?"

"Voltmeter was hanging up around ten thousand volts just before the amp-meter went bye-bye."

"Um-m-m. Ten thousand volts at a hundred amps. That is one million watts, my friends, and no small potatoes. To run the Station's communicating equipment we need seven times that much. Can we do it?"

"We can. I'll have Jim Warren start running the main power bus down here and we'll try it. Meanwhile, we've got a healthy cable from the generator room; we can run the noncommunicating drain of the Station from our plaything here. That should give us an idea. We can use a couple of million watts right there. If this gadget will handle it, we can make one that will take the whole load without groaning. I'm calling Jim right now. He can start taking the load over from the generators as we increase our intake. We'll fade, but not without a flicker."

Walt hooked the output terminals of the tube to the huge cable blocks, using sections of the same heavy cable.

Jim Warren called: "Are you ready?"

"Fade her in," said Walt. He kept one eye on the line voltmeter and opened the anode-coupler slightly. The meter dipped as Warren shunted the Station load over to the tube circuit. Walt brought the line voltage up to above normal, and it immediately dropped as Warren took more load from the solar intake. This jockeying went on for several minutes until Warren called: "You've got it all. Now what?"

"Start running the bus down here to take the communications load," said Don. "We're running off of an eight hundred thousand mile cathode now, and his power output is terrific. Or better, Jim, run us a high-tension line down here and we'll save silver. We can ram ten thousand volts up there for transformation. Get me?"

"What frequency?"

"Yeah," drawled Channing, "have Charley Thomas run us a control line from the primary frequency standard. We'll control our frequency with that."

O.K.?"

"Right-o."

Channing looked at the set-up once more. It was singularly unprepossessing, this conglomeration of iron and steel and plastic. There was absolutely nothing to indicate the two and one third million watts of power that coursed from Sol, through its maze of anodes, and into the electric lines of Venus Equilateral. The cathodes and dynode glowed with their usual dull red glow, but there was no coruscating aura of power around the elements of the system. The gymbals that held the big tube slid easily, permitting the tube to rotate freely as the selsyn motor kept the tube pointing at Sol. The supply cables remained cool and operative, and to all appearances, the set-up was inert.

"O.K., fellows," said Channing. "This is it—"

He was interrupted by the frantic waving of Kingman, from the other side of the air lock.

"I feel slightly conscience-stricken," he said with a smile that showed that he didn't mean it at all. "But let us go and prepare the goat for shearing."

Kingman's trouble was terrific, according to him. "Mr. Channing," he complained, "you are not following our wishes. And you, Mr. Farrell, have been decidedly amiss in your hobnobbing with the engineers here. You were sent out as my consultant, not to assist them in their endeavors."

"What's your grief?" asked Channing.

"I find that your laboratory has been changing the circuits without having previously informed me of the proposed change," complained Kingman. "I feel that I am within my rights in removing the tubes brought here. Your investigations have not been sanctioned—" he looked out through the air lock. "What are you doing out there?"

"We have just succeeded in taking power from the sun," said Don. He tried to keep his voice even, but the exultation was too high in him, and his voice sounded like sheer joy.

"You have been—" Kingman did a double-take. "You *what*?" he yelled.

"Have succeeded in tapping Sol for power."

"Why, that's wonderful."

"Thank you," said Don. "You will no doubt be glad to hear that Wes Farrell was instrumental in this program."

"Then a certain part of the idea is rightfully the property of Terran Electric," said Kingman.

"I am afraid not," said Don. "Dr. Farrell's assistance was not requested. Though his contribution was of great value, it was given freely. He was not solicited. Therefore, since Terran Electric was not consulted formally, Dr. Farrell's contribution to our solar power beam can not be considered as offering a hold on our discovery."

"This is true, Dr. Farrell?"

"I'm afraid so. You see, I saw what was going on and became interested, academically. I naturally offered a few minor suggestions, in somewhat the same manner as a motorist will stop and offer another motorist assistance in changing a tire. The problem was interesting to me and as a problem, it did not seem to me—"

"Your actions in discussing this with members of the Venus Equilateral technical staff without authorization will have cost us plenty," snapped Kingman. "However, we shall deal with you later."

"You know," said Farrell with a cheerfully malicious grin, "if you had been less stuffy about our tubes, they might be less stuffy about my contribution."

"Ah, these nonlegal agreements are never satisfactory. But that is to be discussed later. What do you intend to do with your invention, Dr. Channing?"

Channing smiled in a superior manner. "As you see, the device is small. Yet it handles a couple of million watts. An even smaller unit might be made that would suffice to supply a home, or even a community. As for the other end, I see no reason why the size might not be increased to a point where it may obsolete all existing power-generating stations."

Kingman's complexion turned slightly green. He swallowed hard. "You, of course, would not attempt to put this on the market yourself."

"No?" asked Channing. "I think you'll find that Interplanetary Communications is as large, if not larger, than Terran Electric, and we have an enviable reputation for delivering the goods. We could sell refrigerators to the Titan Colony if we had the V-E label on them and claimed they were indispensable. Our escutcheon is not without its adherents."

"I see," said Kingman. His present volubility would not have talked a jury into freeing the armless wonder from a pickpocketing charge. "Is your invention patentable?"

"I think so. While certain phases of it are like the driver tube, which, of course, is public domain, the applications are quite patentable. I must admit that certain parts are of the power transmission tube, but not enough for you to claim a hold, I know. At any rate, I shall be busy for the next hour, transmitting the details to Washington, so that the Interplanetary Patent Office may rule on it. Our Terran legal department has a direct line there, you know, and they have been directed to maintain that contact at all cost."

"May I use your lines?"

"Certainly. They are public carriers. You will not be restricted any more than any other man. I am certain that our right to transmit company business without waiting for the usual turn will not be contested."

"That sounds like a veiled threat."

"That, sounds like slander!"

"Oh no. Believe me. But wait, Dr. Channing. Is there no way in which we may meet on a common ground?"

"I think so. We want free hand in this tube proposition."

"For which rights you will turn over a nominal interest in solar power?"

"Forty percent."

"But we—"

"I know, you want control."

"We'd like it."

"Sorry. Those are our terms. Take 'em or leave 'em."

"Supposing that we offer you full and unrestricted rights to any or all developments you or we make on the Martian transmission tubes?"

"That might be better to our liking."

"We might buck you," said Kingman, but there was doubt in his voice.

"Yes? You know, Kingman, I'm not too sure that Venus Equilateral wants to play around with power except as a maintenance angle. What if we toss the solar beam to the public domain? That is within our right, too."

Kingman's green color returned, this time accompanied with beads of sweat. He turned to Farrell. "Is there nothing we can do? Is this patentable?"

"No—Yes," grinned Farrell.

Kingman excused himself. He went to the office provided for him and began to send messages to the Terran Electric offices at Chicago. The forty minute wait between message and answer was torture to him, but it was explained to him that light and radio crossed space at one hundred and eighty-six thousand miles per second and that even an Act of Congress could do nothing to hurry it. Meanwhile, Channing's description tied up the Terran Beam for almost an hour at the standard rate of twelve hundred words per minute. Their answers came within a few minutes of one another.

Channing tossed the 'gram before Kingman. "Idea definitely patentable," said the wire.

Kingman stood up. Apparently the lawyer believed that his pronouncement would carry more weight by looming over the smiling, easy-going faces of his parties-of-the-second-part. "I am prepared to negotiate with your legal department; offering them, and you, the full rights to the use of the transmission tube. This will include full access to any and all discoveries, improvements, and/or changes made at any time from its discovery to the termination of this contract, which shall be terminated only by absolute mutual agreement between Terran Electric and Interplanetary Communications.

"In return for this, Interplanetary Communications will permit Terran Electric to exploit the solar beam tube fully and freely, and exclusively—"

"Make that slightly different," said Channing. "Terran Electric's rights shall prevail exclusively—*except* within the realm of space, upon man-made celestial objects, and upon the satellites and minor natural celestial bodies where stations of the Interplanetary Communications Company are established."

Kingman thought that one over. "In other words, if the transport companies desire to use the solar beam, you will hold domain from the time they leave an atmosphere until they again touch—"

"Let's not complicate things," smiled Don cheerfully. "I like uncomplicated things."

Kingman smiled wryly. "I'm sure," he agreed with fine sarcasm. "But I see your point. You intend to power the communications system with the solar beam. That is natural. Also, you feel that a certain amount of revenue should be coming your way. Yes, I believe that our legal departments can agree."

"So let's not make the transport companies change masters in mid-space," smiled Don.

"You are taking a lot on your shoulders," said Kingman. "We wouldn't permit our technicians to dictate the terms of an agreement."

"You are not going to like Venus Equilateral at all," laughed Don. "We wouldn't permit our legal department to dabble in things of which they know nothing. Years ago, when the first concentric beam was invented, which we now use to punch a hole in the Heaviside Layer, Communications was built about a group of engineers. We held the three inner planets together by the seat of our pants, so to speak, and nurtured communications from a slipshod, hope-to-God-it-gets through proposition to a sure thing. Funny, but when people were taking their messages catch as catch can, there was no reason for legal lights. Now that we can and do insure messages against their loss, we find that we are often tangled up with legal red tape."

"Otherwise, we wouldn't have a lawyer on the premises. They serve their purpose, no doubt, but in this gang, the engineers tell the attorneys how to run things. We shall continue to do so. Therefore you are speaking with the proper parties, and once the contract is prepared by you, we shall have an attorney run through the whereases, wherefores, and parties of the first,

second, and third parts to see that there is no sleight of hand in the microscopic type."

"You're taking a chance," warned Kingman. "All men are not as fundamentally honest as Terran Electric."

"Kingman," smiled Channing, "I hate to remind you of this, but who got what just now? We wanted the transmission tube."

"I see your point. But we have a means of getting power out of the sun."

"We have a hunk of that too. It would probably have been a mere matter of time before some bright bird at Terran found the thing as it was."

"I shall see that the contract gives you domain over man-made objects in space—including those that occasionally touch upon the natural celestial objects. Also the necessary equipment operating under the charter of Interplanetary Communications, wherever or whenever it may be, including any future installations."

"Fine."

"You may have trouble understanding our feelings. We are essentially a space-born company, and as such we can have no one at the helm that is not equipped to handle the technical details of operation in space." Channing smiled reminiscently. "We had a so-called efficiency expert running Venus Equilateral a couple of years ago, and the fool nearly wrecked us because he didn't know that the airplant was not a mass of highly complicated, chemical reaction machinery instead of what it really is. Kingman, do you know what an airplant is?"

"Frankly no. I should imagine it is some sort of air-purifying device."

"You'll sit down hard when I tell you that the airplant is just what it is. Martian Sawgrass! What better device in the solar system can be used for air-purifying than a chlorophyll-bearing plant; it takes in carbon dioxide and gives off oxygen. Brother Burbank tossed it in the incinerator because he thought it was just weeds, cluttering up the place. He was allergic to good engineering, anyway."

"That may be good enough in space," said Kingman, "but on Terra, we feel that our engineers are not equipped to dabble in the legal tangles that follow

when they force us to establish precedent by inventing something that has never been covered by a previous decision."

"O.K.," said Don. "Every man to his own scope. Write up your contract, Kingman, and we'll all climb on the bandwagon with our illiterary X's."

In Evanston, North of Chicago, the leaves changed from their riotous green to a somber brown, and fell to lay a blanket over the earth. Snow covered the dead leaves, and Christmas, with its holly went into the past, followed closely by New Year's Eve with its hangover.

And on a roof by the shore of Lake Michigan, a group of men stood in overcoats beside a huge machine that towered above the great letters of the Terran Electric Company sign that could be seen all the way from Gary, Indiana.

It was a beautiful thing, this tube; a far cry from the haywire thing that had brought solar power to Venus Equilateral. It was mounted on gymbals, and the metal was bright-plated and perfectly machined. Purring motors caused the tube to rotate to follow the sun.

"Is she aligned?" asked the project engineer.

"Right on the button."

"Good. We can't miss with this one. There may have been something sour with the rest, but this one ran Venus Equilateral—the whole Relay Station—for ten days without interruption."

He faced the anxious men in overcoats. "Here we go," he said, and his hand closed upon the switch that transferred the big tube from test power to operating power.

The engineer closed the switch, and stepped over to the great, vaned, air-cooled ammeter shunt. On a panel just beyond the shunt the meter hung—

At Zero!

"Um," said the project engineer. "Something wrong, no doubt."

They checked every connection, every possible item in the circuit.

"Nothing wrong."

"Oh now look," said the project engineer, "This isn't hell, where the equipment is always perfect except that it doesn't work."

"This is hell," announced his assistant. "The thing is perfect except that it doesn't work."

"It worked on Venus Equilateral."

"We've changed nothing, and we handled that gadget like it was made of cello-gel. We're running the same kind of voltage, checked on Standard Voltmeters. We're within one tenth of one percent of the original operating conditions. But—no power."

"Call Channing."

The beams between Terra and Venus Equilateral carried furious messages for several hours. Channing's answer said: "I'm curious. Am bringing the experimental ship to Terra to investigate."

The project engineer asked: "Isn't that the job that they hooked up to use solar power for their drive?"

His assistant said: "That's it. And it worked."

"I know. I took a run on it!"

Channing was taking a chance, running the little *Anopheles* to Terra, but he knew his ship, and he was no man to be overcautious. He drove it for Terra at three G, and by dead reckoning, started down into Terra's blanket of air, heading for the Terran Electric plant which was situated on the lake shore.

Then down out of the cloudless sky came the *Anopheles* in a free fall. It screamed with the whistle of tortured air as it fell, and it caught the attention of every man that was working at Terran Electric.

Only those on the roof saw the egg-shaped hull fall out of the sky unchecked; landing fifteen hundred yards off shore in Lake Michigan.

The splash was terrific.

"Channing—!" said the project engineer, aghast.

"No, look, there—a lifeship!"



Cautiously sliding down, a minute lifeship less than the size of a freight car came to a landing in the Terran Electric construction yard. Channing emerged, his face white. He bent down and kissed the steel grille of the construction yard fervently.

Someone ran out and gave Channing a brown bottle. Don nodded, and took a draw of monstrous proportions. He gagged, made a face, and smiled in a very wan manner.

"Thanks," he said shakily. He took another drink, of more gentlemanly size.

"What happened?"

"Dunno. Was coming in at three G. About four hundred miles up, the deceleration just quit. Like that! I made it to the skeeter, here, in just enough time to get her away about two miles ago. *Whoosh!*"

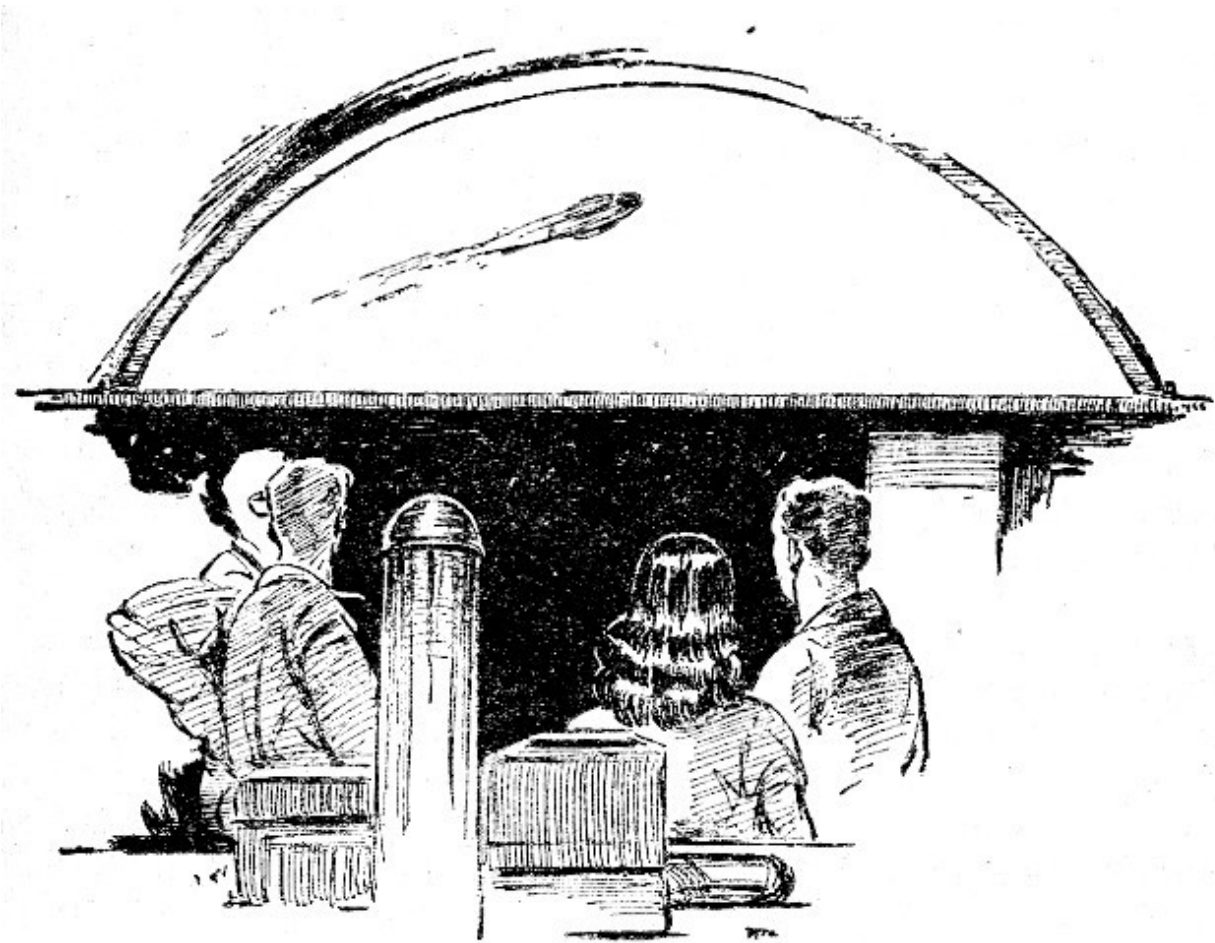
Don dug into his pocket and found cigarettes. He lit up and drew deeply. "Something cock-eyed, here. That stoppage might make me think that my tube failed; but—"

"You suspect that our tube isn't working for the same reason?" finished the project engineer.

"Yes. I'm thinking of the trick, ultra-high powered, concentric beams we have to use to ram a hole through the Heaviside Layer. We start out with three million watts of sheer radio frequency and end up with just enough to make our receivers worth listening to. Suppose this had some sort of Heaviside Layer?"

"In which case, Terran Electric hasn't got solar power," said the project engineer. "Tim, load this bottle into the *Electric Lady*, and we'll see if we can find this barrier." To Channing, he said: "You look as though you could stand a rest. Check into a hotel in Chicago and we'll call you when we're ready to try it out."

Channing agreed. A shave, a bath, and a good night's sleep did wonders for his nerves, as did a large amount of Scotch. He was at Terran Electric in the morning, once more in command of himself.



Up into the sky went the ship that carried the solar tube. It remained inert until the ship passed above three hundred and forty miles. Then the ammeter needle swung over, and the huge shunt grew warm. The tenuous atmosphere outside of the ship was unchanged, yet the beam drew power of gigantic proportions.

They dropped again. The power ceased.

They spent hours rising and falling, charting this unknown barrier that stopped the unknown radiation from bringing solar power right down to earth. It was there, all right, and impervious. Above, megawatts raced through the giant shunt. Below, not even a micro-ammeter could detect a trace of current.

"O.K., Don," said the project engineer. "We'll have to do some more work on it. It's nothing of your doing."

Mark Kingman's face was green again, but he nodded in agreement. "We seem to have a useless job here, but we'll think of something."

Channing left for Venus Equilateral in two more days. They studied the barrier and established its height as a constant three hundred and thirty-nine, point seven six miles above Terra's mythical sea level. It was almost a perfect sphere, that did not change with the night and day as did the Heaviside Layer. There was no way to find out how thick it was, but thickness was of no importance, since it effectively stopped the beam.

And as Don Channing stepped aboard the *Princess of the Sky* to get home again, the project engineer said: "If you don't mind, I think we'll call that one the Channing Layer!"

"Yeah," grinned Don, pleased at the thought, "and forever afterward it will stand as a cinder in the eye of Terran Electric."

"Oh," said the project engineer, "We'll beat the Channing Layer."

The project engineer was a bum prophet—

THE END.

*** END OF THE PROJECT GUTENBERG EBOOK THE LONG WAY

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